Electrical Challenge: BSPD

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1. Rules

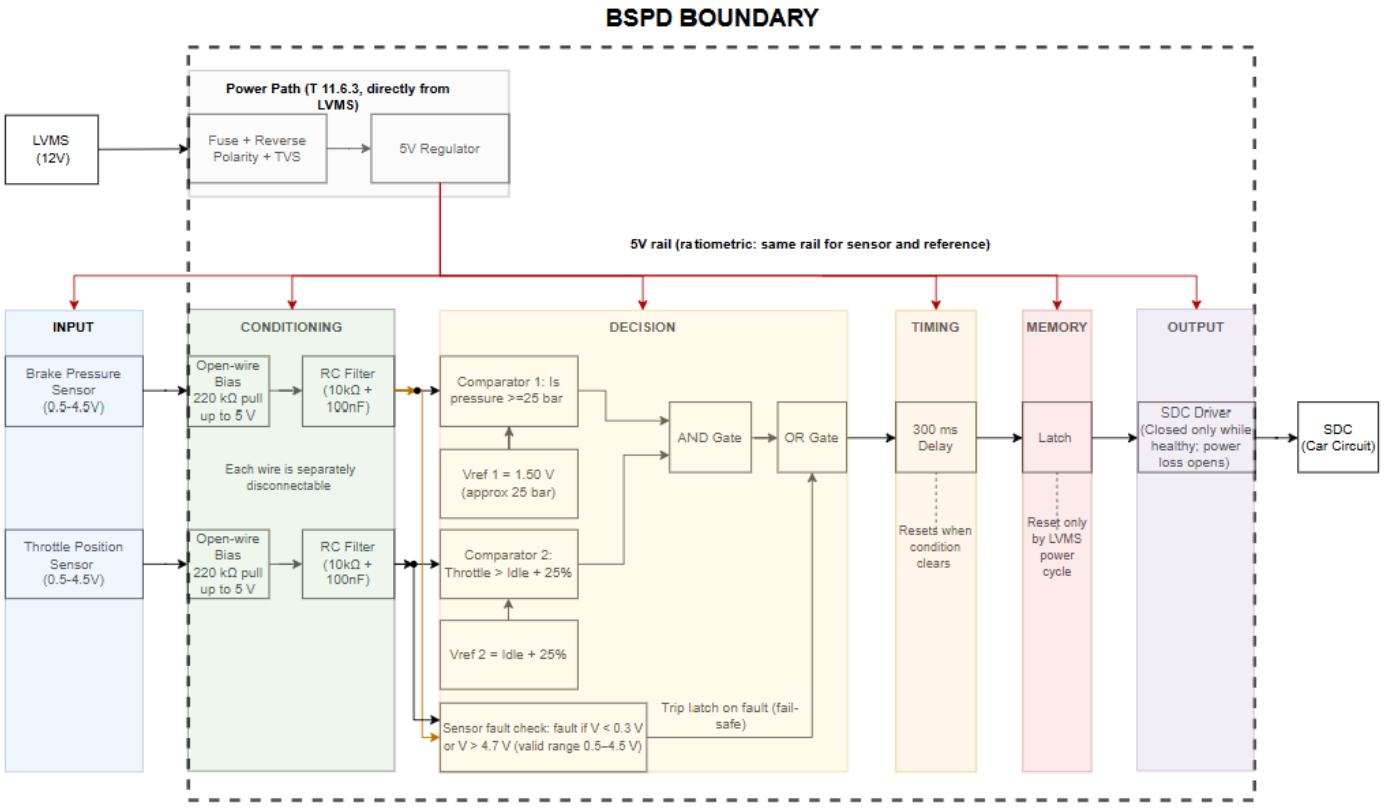
Rule	What it requires	How this design meets it
T 11.6.1	A standalone, non-programmable circuit must open the SDC when hard braking occurs while the throttle is more than 25%.. over idle. The SDC must remain open until the LVMS is power cycled (the master is switched on and off) or the BSPD may self-reset after the condition has been absent for over 10 s.	Only comparators and 74HC logic are used; no microcontroller or firmware detects brake AND throttle conditions. The U6 latch holds the trip until power is removed. The self-reset option is not used. Why? 1. It needs no extra parts 2. Its the safer behaviour 3. Its simple
T 11.6.2	The SDC must open if the implausibility persists for more than 500 ms.	A 300 ms RC timer is used (302 ms nominal; 235–380 ms worst case, including the relay), remaining below 500 ms. accommodating the rough +-25% variation.
T 11.6.3, T 1.3.1, T 11.3	The BSPD must be supplied directly from the LVMS. The connection must not pass through a common PCB or any device other than overcurrent protection or connectors.	One dedicated 12 V wire runs from the LVMS output (T 11.3) to connector J1, passing only through the connector. Polyfuse F1 on the BSPD board provides overcurrent protection. Switching off the LVMS removes power and resets the latch.
T 11.6.4	No additional functionality is permitted. Interfaces are limited to power, required sensors and the SDC. Sensor and supply wires must not pass through other devices first. If sensors are shared, interference must be ruled out.	Only three interfaces are provided: J1 (power), J2/J4 (sensors) and J6 (SDC). There are no data, CAN or indicator outputs. Sensor wires run directly to the board.

T 11.6.5	Hard braking must be detected using a brake-pressure sensor. The threshold must avoid wheel lock-up and must be no greater than 30 bar.	A Variohm EPT3100 pressure sensor is used, with a 1.50 V threshold corresponding to approximately 25 bar. The threshold remains below 30 bar with tolerances.
T 11.6.7	Each sensor signal wire must be separately disconnectable for technical inspection.	Each sensor signal has its own removable link (J3 for brake and J5 for throttle) and screw terminal, so pulling a link does it without any tools
T 11.6.8, T 11.9.1–2, T 11.9.5	All BSPD signals are System Critical Signals. Open circuits, shorts to ground and analogue shorts to supply must lead to the safe state. For a CV, this means opening the SDC and stopping the engine.	Window comparators flag voltages below 0.30 V and above 4.70 V, outside the valid 0.5–4.5 V range. A pull-up resistor makes a disconnected wire read about 5 V, so it's caught. That covers a broken wire, a short to ground and a short to supply. All fault,s feed the same 300 ms timer and latch. The safe state is the relay released, the shutdown circuit open, and the car stopped
CV 4.1.2, CV 4.1.3	The BSPD must be a series element of the SDC. SDC circuits must open when de-energised or disconnected.	A normally open K1 contact is connected in series with the SDC. Its coil is energised only while the system is healthy. Loss of supply, a blown fuse, an unplugged J1 or a failed driver causes the SDC to open.
IN 4.1.3	BSPD operation must be checked during Electrical Inspection.	Links J3/J5 and the sensor terminals allow the inspector to inject test voltages or open each signal, following the test plan in Section 6.2.

2. Overview

The rules (T 11.6, T 11.9, CV 4.1) turn the BSPD into three requirements: detect “hard braking while the throttle is more than 25% open”, open the shutdown circuit if that persists beyond 500 ms (and stay open until the LVMS is cycled), and treat any sensor wiring fault as a safe-state event. I interpreted the challenge as designing a fail-safe with no microcontroller. The relay is held closed only while everything is healthy, so a trip, a fault or a loss of power opens the SDC. My process: turn the rules into requirements; draw a block diagram to interpret the logic visually; set thresholds from the sensor curves (brake: V = 0.5 + 0.04·P); design each stage in KiCad with real part numbers and calculate thresholds and timing; check against each rule; draw a draft two-layer PCB; and plan how to build, test and fit it to the car.

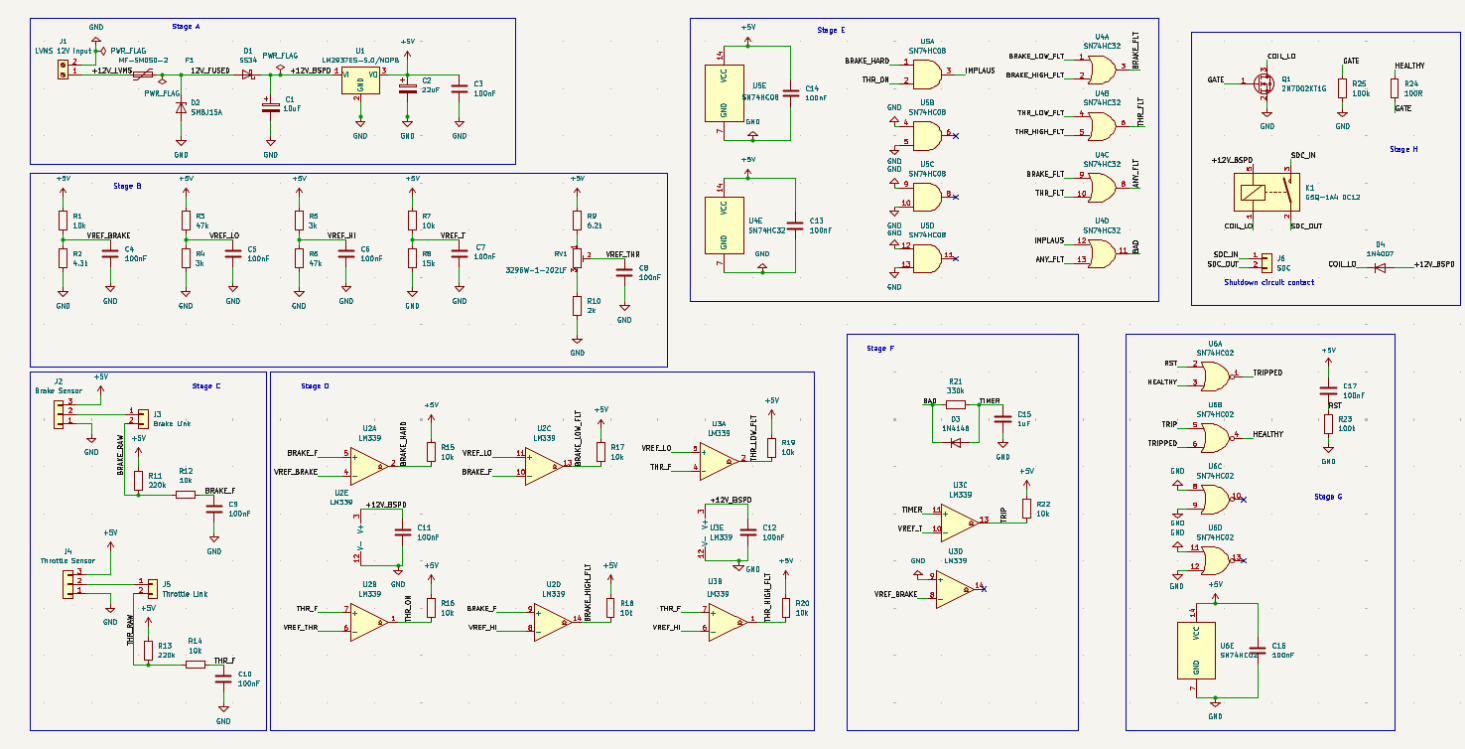
3. Logic/block diagram



1. **Power:**
    a. The circuit takes its power directly from the car's main switch (LVMS, 12V), as the rules require (T11.6.3).
    b. A fuse and reverse polarity act as protection guards against faults and a wrongly connected battery, where a TVS diode absorbs voltage spikes.

   c. A 5V regulator then turns the 12V into 5V so it can power the rail that feeds the system.
2. **Input (two analogue voltages):**
   a. Brake pressure (0.5-4.5 V for 0-100 bar)
   b. Throttle position (assumed 0.5–4.5 V ratiometric, which can be confirmed from the TPS280 datasheet).
3. **Conditioning**
   a. Each passes through a removable link (T 11.6.7).
   b. a 220 kΩ pull-up to 5 V (a broken wire reads 5 V)
   c. 10 kΩ/100 nF low-pass filter (159 Hz) to smooth out electrical ignition noise.
4. **Decision:**
   a. Comparator U2A indicates hard braking (BRAKE_HARD) when the brake voltage exceeds 1.50 V (25 bar).
   b. Comparator U2B indicates THR_ON when the throttle voltage exceeds VREF_THR, a set level for idle + 25%
   c. An AND gate (U5A) outputs IMPLAUS when both of its inputs are on.
   d. Sensor-fault detection (T 11.9):
      i. Each sensor has two comparators, allowing it to form a window: below 0.30 V (short to ground) or above 4.70 V (short to supply or open).
      ii. The four fault outputs are combined by OR gates.
5. **Timing:**
   a. The fault must last 300 ms before the circuit reacts.
      i. This ignores the brief blips (driver heel and toe breaking).
      ii. Well within the 500 ms rule set (T 11.6.2)
6. **Latch memory:**
   a. Built from two NOR gates, the latch remembers the fault and holds it.
   b. It stays set even after the indicators are cleared and can only be reset by removing power (LVMS power cycle).
7. **Output:**
   a. This is the relay that closes the shutdown circuit only when everything is healthy.
   b. If the fault trips or the power is lost, the relay opens, ensuring that every failure ends in a safe state.
   c. The SDC is the car's chain of safety switches essentially.

4.Schematic Design



Stage A: Supply, protection and 5V

- LVMS 12 V enters on J1 and meets, in order:
  - F1 (MF-SM050-2 resettable fuse, 0.5 A hold, 60 V) for overcurrent protection;
  - D2 (SMBJ15A TVS) clamping surges and load-dump spikes;
  - D1 (SS34 Schottky) blocking reverse polarity
  - C1 (10 µF) as bulk storage.
- The result is the +12V_BSPD rail, which feeds the comparators and the relay coil.
- U1 (LM2937ES-5.0, 500 mA low-dropout regulator) makes +5 V for the sensors, the reference dividers, the pull-ups and the logic.
- C2 (22 µF tantalum) is chosen because the LM2937 datasheet requires at least 10 µF with an ESR between 10 mΩ and 3 Ω over the full temperature range, which an aluminium electrolytic cannot guarantee when cold; C3 adds 100 nF for high frequencies.
- Because the reference voltages and the sensors share this 5 V rail, any drift in the supply moves the thresholds together with the sensor signals.

Stage B: Reference voltages

- All thresholds are resistor dividers from +5 V with 100 nF bypass capacitors (C4-C8).
- VREF_BRAKE = 5 × 4.3 k / (10 k + 4.3 k) = 1.50 V, which is 25 bar (V = 0.5 + 0.04·P).
- VREF_LO = 5 × 3 k / 50 k = 0.30 V
- VREF_HI = 5 × 47 k / 50 k = 4.70 V bound the valid sensor range.

- VREF_T = 5 × 15 k / 25 k = 3.00 V sets the timer.
- VREF_THR is the throttle threshold: R9 (6.2 k) + trimmer RV1 (2 k, 3296W) + R10 (2 k) gives an adjustable 0.98-1.96 V, which covers “25 % over idle” for a range of idle voltages (for example, idle = 0.5 V gives 1.5 V).

Stage C: Sensor inputs

- J2 (brake) and J4 (throttle) are 3-pin terminals: GND, signal and +5 V.
- The signal passes through link J3 / J5 (the T 11.6.7 disconnect point)
- It then meets a 220 kΩ pull-up (R11/R13) to +5 V
- then a 10 kΩ + 100 nF low-pass filter (R12/C9, R14/C10; f= 159 Hz).
- If a wire is cut or a link removed, the pull-up pulls the node to 5 V, above the 4.70 V limit, which is flagged as a fault.
    - The 220 kΩ pull-up has negligible effect on a normal sensor (for a source impedance up to about 1 kΩ the error is below 20 mV).

Stage D: Comparators

- Two LM339 quad comparators (U2, U3) have open-collector outputs, each pulled up to 5 V by 10 kΩ (R15-R20, R22).
    - U2A (BRAKE_F above VREF_BRAKE) gives BRAKE_HARD
    - U2B (THR_F above VREF_THR) gives THR_ON.
    - U2C/U2D and U3A/U3B are the window comparators giving BRAKE_LOW_FLT, BRAKE_HIGH_FLT, THR_LOW_FLT and THR_HIGH_FLT.
- The comparators are powered from +12V_BSPD (U2E/U3E, decoupled by C11/C12), not from 5 V, because the LM339 input common-mode range only reaches about VCC-1.5 V; on a 5 V supply the 4.7 V reference would be outside it.
    - The 5 V pull-ups keep outputs at logic levels for the 74HC gates.

Stage E: Combinational logic

- U5A (74HC08 AND): IMPLAUS = BRAKE_HARD AND THR_ON.
- U4A/U4B (74HC32 OR) combine each sensor’s low and high fault into BRAKE_FLT and THR_FLT
- U4C gives ANY_FLT
- U4D gives BAD = IMPLAUS OR ANY_FLT
- Unused gates have their inputs tied to ground and floating outputs so they cannot oscillate.
- C13/C14 decouple the supply pins.

Stage F: 300 ms persistence timer

- BAD charges C15 (1 µF) through R21 (330 kΩ) towards 5 V.
- Comparator U3C raises TRIP when the TIMER node passes VREF_T = 3.00 V:
- D3 (1N4148) across R21 discharges C15 quickly when BAD goes low, so every event starts the timing

Stage G: latch and power-on reset

- U6A and U6B are cross coupled NOR gates:
    - TRIPPED = NOR(RST, HEALTHY) and HEALTHY = NOR(TRIP, TRIPPED).

- TRIP sets TRIPPED; it stays set until power is removed, which satisfies T 11.6.1 (SDC stays open until the LVMS is power-cycled).
- At power-up, C17 (100 nF) and R23 (100 kΩ) give a 10 ms pulse on RST, forcing the latch to the healthy state so the car can start after a power cycle.
- Unused gates U6C/U6D are parked with grounded inputs.

Stage H: Relay driver and SDC contact

- HEALTHY drives the gate of Q1 (2N7002K, logic-level N-MOSFET) through R24 (100 Ω)
- R25 (100 kΩ) holds the gate low if the drive is lost.
- Q1 switches the coil of K1 (Omron G5Q-1A4 DC12: 12 V, 720 Ω, 16.7 mA) between +12V_BSPD and ground.
- D4 (1N4007) is the flyback diode that absorbs the coil's inductive spike when Q1 turns off.
- K1's normally open contact connects SDC_IN to SDC_OUT on J6, in series with the rest of the SDC.
    - The coil is energised only while HEALTHY is high, so the circuit is fail-safe
    - if power, the fuse, the driver or the logic fails, the relay releases and the SDC opens (CV 4.1.3).

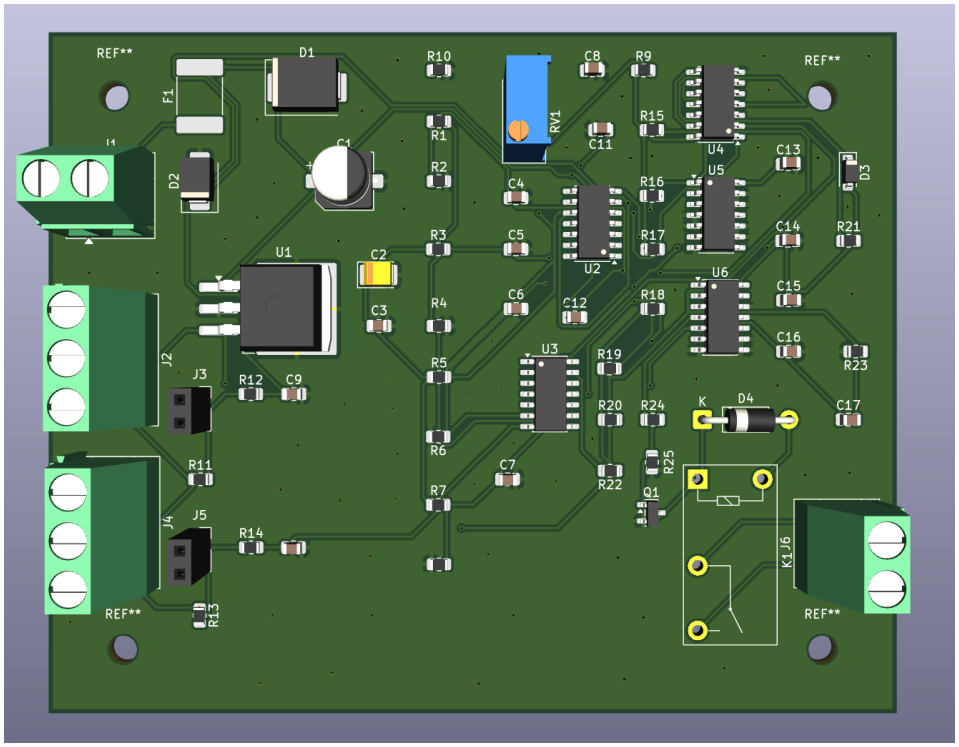
Part numbers and reason for choice

Ref	Part number	Role	Why I chose it
U1	LM2937ES-5.0/NOPB	5 V regulator	It's made for cars, so it can handle voltage spikes and reverse voltage on a 12 V system. It gives 500 mA, which is way more than I need, and the big tab on the TO-263 package spreads the heat into the copper
U2, U3	LM339	Quad comparator	It runs from a single supply (up to 36 V), so I can power it from 12 V. Its open-collector outputs can still drive 5 V logic. It's also cheap and easy to get
U4	SN74HC32	Quad OR gate	Standard logic chip with no programming involved, which the rules need. The SOIC-14 package is big enough to solder by hand
U5	SN74HC08	Quad AND gate	Same reasons as U4
U6	SN74HC02	Quad NOR gate	Two of the NOR gates connected together make the SR latch that remembers a trip

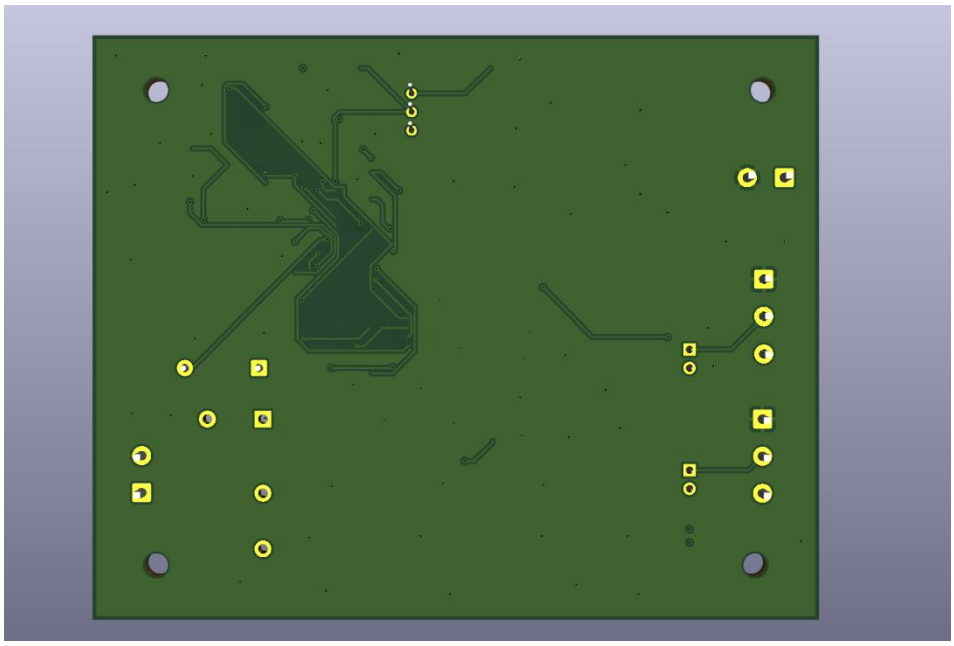
Q1	2N7002KT1G	Relay coil driver	A logic-level 60 V MOSFET with a gate that's protected from static. It needs much less drive current than a normal transistor
K1	G5Q-1A4 DC12 (Omron)	SDC relay	Sealed 12 V relay that only draws 16.7 mA. It's normally open, so it opens when it loses power, which is what CV 4.1.3 asks for.
F1	MF-SM050-2 (Bourns)	Resettable fuse	Holds 0.5 A and is rated 60 V. It resets itself, so I don't have to replace it after a fault, and it's fine for my load of under 100 mA
D1	SS34	Reverse-polarity protection	3 A Schottky diode with a small voltage drop (about 0.4 V), so the regulator still has enough input voltage to work
D2	SMBJ15A	TVS diode	Soaks the load-dump and switching spikes before they reach the regulator and capacitors
D3 / D4	1N4148 / 1N4007	Timer reset / relay flyback	The 1N4148 is a fast small diode, so it empties the timer capacitor C15 quickly. The 1N4007 is tougher, so it can handle the voltage spike from the relay coil
RV1	3296W-1-202LF (Bourns)	2 kΩ trimmer	It's multi-turn, so I can set VREF_THR very precisely
J1-J6	5.08 mm screw terminals; 2.54 mm headers (J3, J5)	Connectors	Screw terminals mean no special tools are needed for wiring. The jumper links make it easy to show each sensor wire can be disconnected separately (T 11.6.7)
R, C	1 % 0805 resistors; X7R 0805 capacitors; C1 SMD electrolytic 35 V; C2 tantalum	Passives	Common, easy to hand-solder, and accurate enough to keep the timing well inside the 500 ms limit.

5.PCB Layout

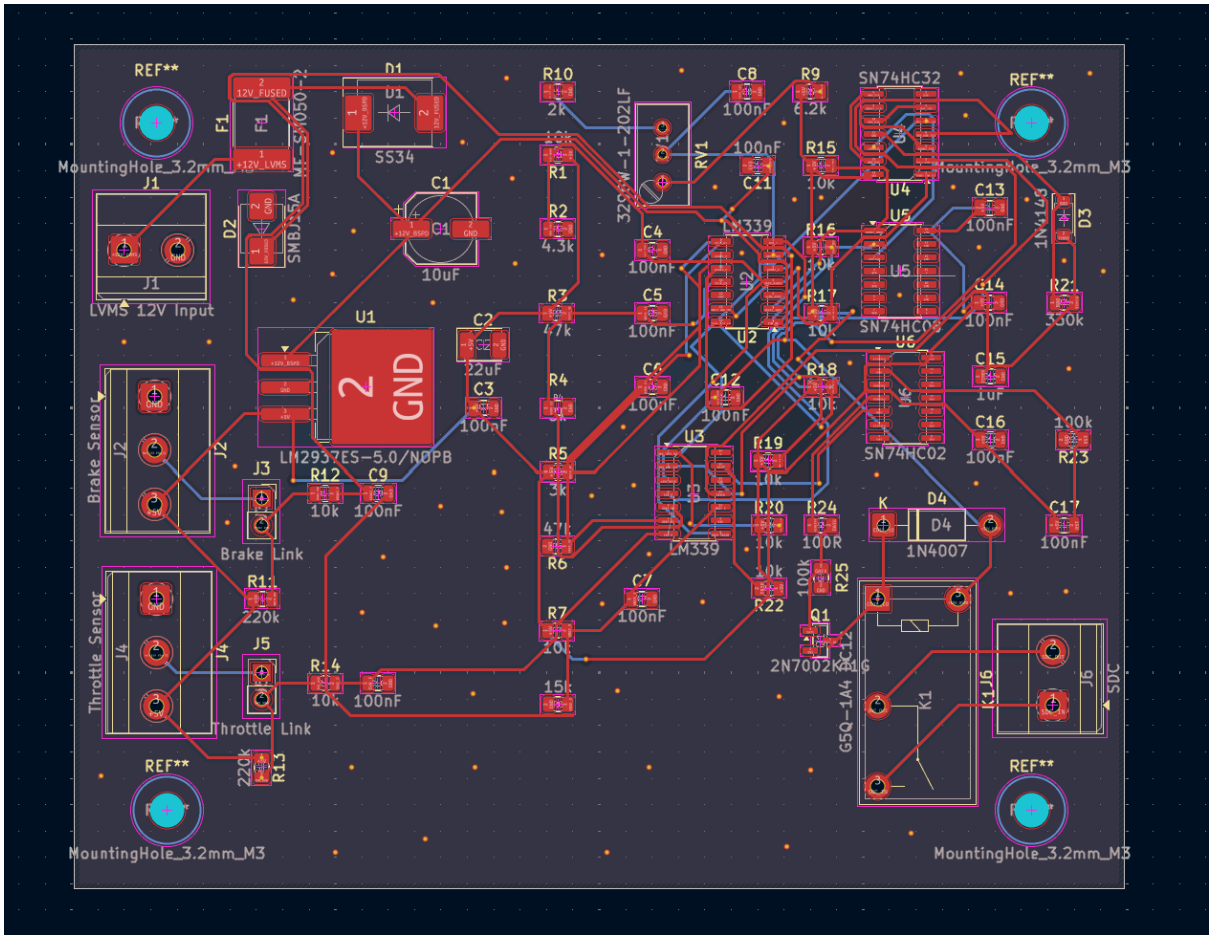
Top View



Bottom View



2D View



Key Decisions

Decision	Reason
Placement follows signal flow	The power chain sits top-left (J1 then F1 then D2 then D1 then C1 then U1), the sensor connectors are on the left edge, the comparators and logic are in the middle, and the relay is next to the SDC connector. **Short tracks keep the sensitive analogue signals quiet.**
100 nF capacitor at each IC supply pin	It supplies the fast bursts of current right next to the chip, which stops glitches that could cause a false trip.
Net classes: 0.8 mm for 12 V/5 V, 1.0 mm for the SDC, 0.25 mm for signals	Power and SDC tracks carry much more current than signals, so they need to be wider. That way they won't overheat or burn out if there's a fault.

Ground copper pour on both layers, stitched with vias	It gives the current an easy way back to ground, which cuts noise and makes routing easier. The thermal-relief pads keep the board easy to hand-solder.
Four M3 holes, non-plated, 5 mm from the corners	They hold the board firmly so it doesn't move with vibration. They're non-plated so the circuit ground isn't connected to the chassis, which avoids ground loops.
0805 passives, SOIC-14 ICs, 5.08 mm screw terminals	Everything is big enough to solder by hand and to check by eye at Technical Inspection. It's also easy to swap parts at an event.
Relay and terminals are through-hole	Through-hole joints are much stronger, so they cope better with cables being pulled and the car shaking.

6. Implementation

1. Finish and double-check the design.
First I'd finish routing the board and get the design rule check clean. Then I'd go back through the rules (T 11.6) line by line and tick off each one against my schematic. I'd also ask someone else on the team to look it over, because I'll miss mistakes in my own work.

2. Test the risky bits before making a board.
The parts I'm least sure about are the 300 ms timer and the latch, so (ideally) I'd breadboard those first. I'd check on an oscilloscope that the timer really takes about 300 ms and that the latch holds until I power-cycle it. **It's much cheaper to find a mistake here than after the boards are made.**

3. Order the boards and parts.
Before ordering anything, I'd ask the team and the university whether there's a workshop that can make or help with prototype boards, since that would be fastest and cheapest for a first test version. However, for the final board, I'd use a professional manufacturer. I'd order a small batch of PCBs from a cheap manufacturer and parts from a supplier. I'd also order a few extra boards for spares and mistakes. My guess is the parts cost roughly £20 to £30 per board, and the main costs are the relay, the screw terminals and the regulator. I'd check real prices when ordering.

4. Build it in stages.
I'd solder the power section first, with no chips fitted, and check I get a clean 5 V before anything else goes on. This stops a mistake in the power circuit from damaging all the chips. Then I'd add the comparators, then the logic, then the relay.

5. Test it on the bench.
I'd use a bench power supply and two potentiometers to pretend to be the sensors. Then I'd run through every situation I can think of:

- brake only, which should do nothing
- throttle only, which should do nothing

- both for less than 300 ms, which should do nothing
- both for longer than 300 ms, which should trip it
- unplugging each sensor wire
- shorting each sensor wire to ground and to 5 V
- switching the power off and on, which should reset it

I'd also time how long it takes to trip, and I'd test it with the supply between about 9 V and 14 V, because the car's voltage dips when the engine cranks.

6. Set it up with the real sensors.
I assumed the sensors give 0.5 to 4.5 V, so I'd check this with a multimeter on the real ones. Then I'd read the throttle voltage at idle and use the trimmer (RV1) to set the throttle threshold to 25 % above that.

7. Make it survive the car.
A car shakes and gets hot. I'd put the board in a small enclosure, add strain relief on the cables, and mount it away from heat. Basically trying to replicate the environment.

8. Fit it and prove it works.
I'd connect the 12 V straight from the LVMS and put the relay in series with the shutdown circuit. Before inspection I'd test it on the car myself by pressing both pedals, so there are no surprises in front of the scrutineers. I'd also keep a spare board and a test log so anyone can check what I did.

What I'd improve next: use a film capacitor for the timer, since the ceramic one is the least accurate part, and add a bit of hysteresis to the comparators so a sensor sitting right on the threshold can't make the timer restart over and over.